Step 2a - H9 - Electrified mineral drying

Market scan

Market size

The global industrial drying equipment market is ~$5-10B/yr depending on scope (FactMR ~$4.8B 2025; GrandView ~$8.2B 2025; SNS ~$14B 2023), with rotary dryers ~$1.3-4B of that and minerals/ores ~24% of rotary dryer demand. So the mineral drying equipment+services pool relevant here is order-of-magnitude ~$1-3B/yr globally. The underlying product markets are much larger (kaolin ~46 Mt/yr, ~$5-8B; GCC ~$12-21B; ore concentrates far bigger), and drying consumes 10-25% of industrial energy — mostly fossil-fired today. Electrification is at pilot/early-reference stage (Mondo Minerals talc 5 MW electric heater, Nordic Mining electric dryers, Fraunhofer SHS+MVR), so the 'electrified mineral drying' market itself is emerging, worth low $ hundreds of millions and policy-driven.

Region	Size ($B)	Region	Size ($B)
Europe	~$0.3-0.8B (~20% of mineral dryer spend; strongest electrification pull from ETS/gas prices)	Asia	~$1-2B (~40-48% share, China/India; mostly new-build, weaker electrification economics)
America	~$0.5-1.2B (~25-30%; US IRA/DOE funding supports electrification pilots)	Africa	~$0.1-0.3B (~5-10%, incl. Middle East; mining-driven, nascent)

Target customers

A few thousand thermal-drying plants worldwide: Europe alone has ~750 processing plants and 685 mines/quarries among IMA-Europe's 250 member companies (180 Mt/yr minerals). Decision-making is concentrated in majors — Omya runs ~160 plants, Imerys ~100+ sites (48 for CaCO3 alone) — so a handful of corporate engineering/sustainability teams controls a large share of the reachable base; below that sits a long tail of regional and single-quarry operators that are cheap to ignore and hard to sell to. Ore concentrators (copper, iron, nickel laterite, rutile/garnet) add several hundred single-line drying assets with high per-site value. Buyers respond to carbon cost (EU ETS), gas-price volatility, and net-zero commitments; procurement cycles are slow (12-24 months) and references matter.

Customer size	Approx. number	Market share (in %)
Large	~30-50 producer groups worldwide, ~500-800 drying plants (~)	~50-60% of dried mineral tonnage (~)
Medium	~300-500 regional producers, ~1,000-2,000 plants (~)	~30% of tonnage (~)
Small	~1,000-3,000 single-site quarry/processors (~)	~10-15% of tonnage (~)

Players offering a solution today

Incumbent thermal dryer supply is moderately concentrated at the top (top-5 rotary-dryer OEMs ~35-52% of revenue depending on report) but fragmented below, with strong regional and niche OEMs plus aggressive Chinese suppliers. None of the large OEMs leads in electric/microwave drying — the dielectric niche is a field of SMEs (SAIREM, Pueschner, Cellencor, Ferrite/IMS, Muegge, Fricke & Mallah) offering 6-300 kW modules, and utilities/ESCOs (e.g. Nevel) are entering electrification-as-a-service. Whitespace:

no dominant MW-scale electric dryer vendor for bulk minerals yet.

Player	Type of solution	Market share (in %)
Large (Andritz, FLSmidth, Metso, ThyssenKrupp, GEA)	turnkey rotary/fluid-bed/flash dryer lines integrated with grinding, mostly gas/oil-fired; growing service/efficiency-retrofit offers	~35-50% of new mineral dryer revenue (~)
Medium (Allgaier, Comessa, Schenck, Carrier Vibrating, FEECO, Hazemag, Mitchell, Louisville, Buehler, Chinese OEMs)	standard and custom rotary/FBD dryers, regional service, price-competitive	~30-40% (~)
Small electric/dielectric niche (SAIREM, Pueschner, Cellencor, Ferrite/IMS, Muegge; ESCOs like Nevel)	kW-to-300 kW microwave/RF dryers, electric air/SHS heating, electrification projects	<5-10% today, emerging (~)

Regulation

Regulation	Jurisdiction	What it means for us
EU ETS (Directive 2003/87/EC, reformed 2023)	EU	Dryers are explicitly listed combustion units in Annex I; installations >20 MWth pay per tCO2 (~EUR60-90/t recently). Electrifying drying removes the exposure directly.
CBAM + free-allocation phase-out 2026-2034	EU	Free allowances shrink for CBAM sectors and benchmarks tighten elsewhere, raising the effective carbon cost of gas-fired drying for EU mineral producers.
Medium Combustion Plants Directive 2015/2193 & IED	EU	Gas burners of 1-50 MW need permits and NOx/SOx/dust limits; electric heating exits combustion permitting entirely, simplifying new lines.
EU 2040 climate target / Net-Zero Industry Act	EU	Communication targets ~80% fossil-fuel cut by 2040; process-heat electrification is a named lever and benefits from NZIA/Innovation Fund support.
Energy Efficiency Directive (EU) 2023/1791	EU	Mandatory audits/energy-management systems flag drying (10-25% of industrial energy) as a priority action at every large mineral site.

National & US incentives (Germany EEW/carbon contracts, EU Innovation Fund; US IRA 48C, DOE industrial demos)	EU, DE, US	Subsidies and contracts-for-difference specifically fund switching fossil process heat to electric — de-risks customer capex for an electric dryer.

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	
Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis